Alexander D. Kaiser

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Summary of Qualifications

- Computational scientist and applied mathematician with broad expertise in modeling and simulation, including fluid-structure interaction, cardiovascular biomechanics, and development of new modeling methods.
- Technical lead and collaborator with engineering, clinical, and computer science teams in academia, government, and industry. Founded and lead a funded research team at Stanford.
- Mathematician by training working closely with cardiac surgeons and cardiologists to build models around clinical questions. Brought simulation-guided design of surgical repair into clinical use.
- Further projects in Monte Carlo methods, parallel computing, and symbolic simplification.

Education

New York University, Courant Institute of Mathematical Sciences
PhD Mathematics. Thesis: Modeling the Mitral Valve 2017
MS Mathematics. Thesis: Computational Experiments in Markov Chain Monte Carlo 2013

University of California, Berkeley
BA Mathematics, minor in Computer Science 2009

Skills

Methods: mathematical modeling and new method development, computational fluid dynamics, fluid-structure interaction, immersed boundary methods, image-based patient-specific modeling, reduced-order modeling, parameter estimation, numerical methods for differential equations, high-performance computing
Languages: MATLAB, Python; working knowledge of C++, C, and Mathematica
Tools: SimVascular, 3D Slicer, Meshmixer; ParaView, VisIt, PyVista; Slurm, Linux, Git

Experience

Stanford University
Instructor (full-time research role), Cardiothoracic Surgery 2024 –
Research Engineer, Pediatrics (Cardiology) 2022 – 2024
Postdoctoral Scholar, Pediatrics (Cardiology), 2017 – 2022
Institute for Computational & Mathematical Engineering (ICME)

- Founded and lead a research team modeling heart valves, congenital heart disease, and surgical repair. Secured $1.24M in competitive funding as principal investigator from NIH, the American Heart Association, and Stanford.
- Translated simulations to the clinic (Stanford Children's news). The first five pediatric patients with simulation-guided repair had excellent operative outcomes and more consistent valve gradients than historical controls.
- Supervise two direct reports, a postdoctoral scholar and a medical student, and co-mentor two other trainees.
- Extended elasticity-based design to bicuspid aortic and quadricuspid truncal valves for physiology studies.
- Developed simulation-guided design tools for surgical planning of valve repair using fluid-structure interaction.
- Compared full 3D, time-resolved simulated velocity fields against in vitro 4D flow MRI data, the first reported heart valve study to compute relative error over the entire flow field.
- Showed with simulations that, contrary to conventional wisdom, valve performance improves after neonatal truncus arteriosus repair without direct leaflet intervention, a trend confirmed on clinical imaging.

New York University, Courant Institute of Mathematical Sciences
Graduate Student Researcher, Department of Mathematics 2011 – 2017

- Created elasticity-based design, a nearly first-principles method that derives mitral valve geometry, material properties, and fiber orientations from mechanical equilibrium with only limited experimental inputs.
- Performed immersed boundary fluid-structure interaction simulations of the mitral valve that opened freely and closed competently across multiple cardiac cycles.
- Parallelized affine-invariant MCMC samplers on GPUs for parameter estimation in differential equations.

Innovein Medical
Consultant, Medical Device Design 2018 – 2019

- Used fluid-structure interaction simulations to design a new feature for a prosthetic vein valve, removing a major failure mode. The feature was adopted in later designs, and the device entered a clinical feasibility trial.

Lawrence Berkeley National Laboratory
Computer Systems Engineer, Complex Systems & Future Technologies Groups 2009 – 2011

- Led development of a testbed of reference kernels for parallel computing research, published as first author.
- Maintained arbitrary- and extended-precision arithmetic libraries ARPREC and QD (C++), with user support.

University of Newcastle, Australia
Visiting Researcher, Centre for Computer-Assisted Research Mathematics & its Applications 2011

- Designed algorithms for automated simplification of large symbolic sums, released open-source as SimplifySum.
- Achieved over 1000x speedup compared to built-in algorithms in Mathematica on certain problems.

Apple Inc.
Cooperative Researcher, Advanced Computation Group 2010

- Developed a C/Fortran interface for MatrixFFT, Apple's high-performance library for Fast Fourier Transforms.

Jet Propulsion Laboratory, NASA
Summer Student Researcher, Information Processing Group 2007

- Designed and validated schemes that greatly reduced undetected errors (corrupted data passing as correct) in error-correcting codes for the NASA Deep Space Network standard, published as first author.

Funding and Honors

- NIH K25 Mentored Quantitative Research Development Award, $908,880 2024 – 2029
- American Heart Association Career Development Award, $231,000 2024 – 2027
- Stanford Maternal & Child Health Research Institute Instructor K Award Support, $100,000 2025 – 2027
- Kurt O. Friedrichs Prize for Outstanding Dissertation in Mathematics, NYU 2018
- Math Master's Thesis Prize, NYU 2014
- NIH T32 Training Fellowship, Mechanisms and Innovation in Cardiovascular Disease 2018
- National Science Foundation Graduate Research Fellowship 2013

Software

- Primary author of open-source scientific software packages:
  - `heart_valves`: Model generation with elasticity-based design and fluid-structure interaction for aortic and mitral valves (MATLAB, Python, C++)
  - `mc_stretch`: Affine-invariant, GPU-parallel MCMC sampler (C, OpenCL; 2013)
  - `SimplifySum`: Automatic simplification of symbolic sums (Mathematica; 2013)
- Contributed to IBAMR (fluid-structure interaction), svZeroDSolver (lumped-parameter blood flow), svFSI (multiphysics solver), SimVascular (patient-specific modeling), ARPREC and QD (2009 – 2011), MatrixFFT (2010)

Selected Highlight Publications

- Bornemann KM, ..., **Kaiser AD**. Simulations Predict Improved Valve Performance Without Direct Leaflet Intervention After Neonatal Truncus Arteriosus Repair. *J Thorac Cardiovasc Surg*, 2026. arXiv movies
- **Kaiser AD**, ..., Ma MR. Simulation-Based Design of Bicuspidization of the Aortic Valve. *J Thorac Cardiovasc Surg*, 2024. arXiv movies
- **Kaiser AD**, ..., Peskin CS. Modeling the Mitral Valve. *Int J Numer Method Biomed Eng*, 2019. arXiv movies
- Choi PS, **Kaiser AD**, ..., Ma MR. Simulation-Guided Outlet Valve Repair: Early Integration of Computationally-Derived Principles of Free-Edge Length in Bicuspidization Repair. *Front Cardiovasc Med*, 2026.
- Bailey DH, Borwein JM, **Kaiser AD**. Automated Simplification of Large Symbolic Expressions. (alphabetical order per mathematics convention, Kaiser primary author) *J Symb Comput*, 2014.

19 peer-reviewed publications, 6 submitted. Reviewer for 11 journals. Full list in CV.